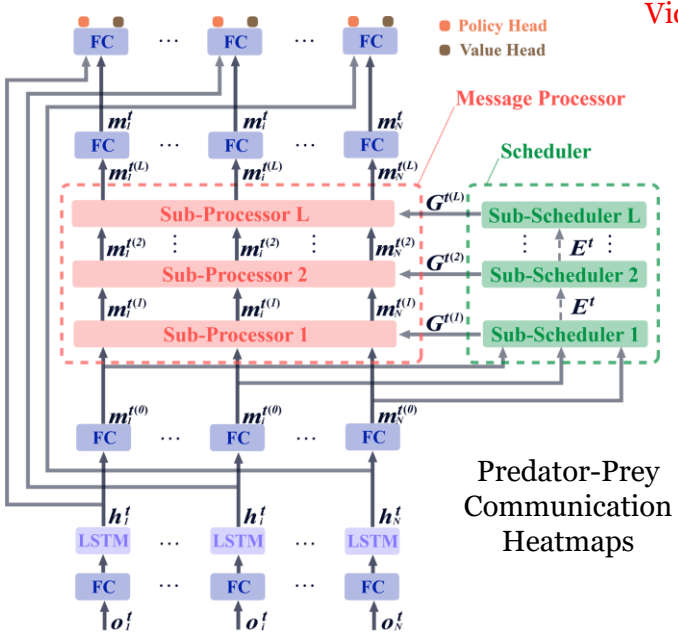
Multi-Agent Reinforcement Learning

Yaru Niu (https://www.yaruniu.com/)

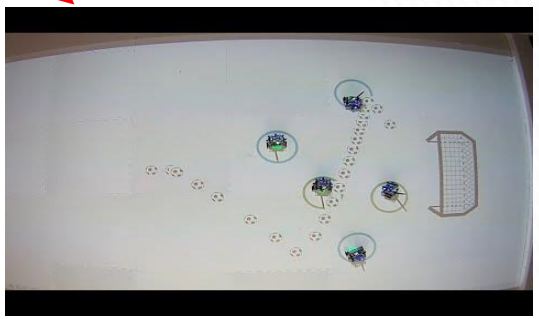
MAGIC: Multi-Agent Graph-AttentIon Communication

- Developed a novel communication protocol for MARL, to solve the problems of when to communicate, whom to address messages, and how to process messages while establishing hierarchical relations between agents.
- Outperformed prior methods across three domains, including the Google Research Football environment, achieving a 10.5% increase in reward. Further, MAGIC learned to communicate 27.4% more efficiently than the average baseline.
- Demonstrated our algorithm on physical robots in a 3-vs.-2 soccer scenario on the Robotarium platform, a physical, multi-robot testbed.

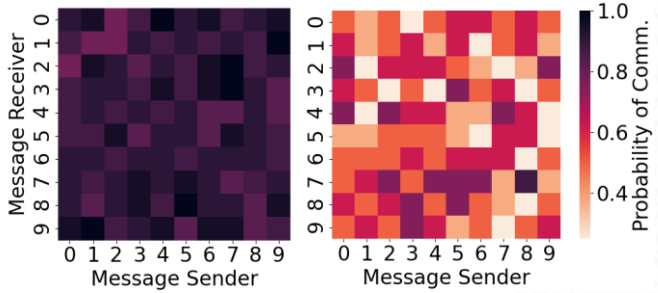


Video, click here

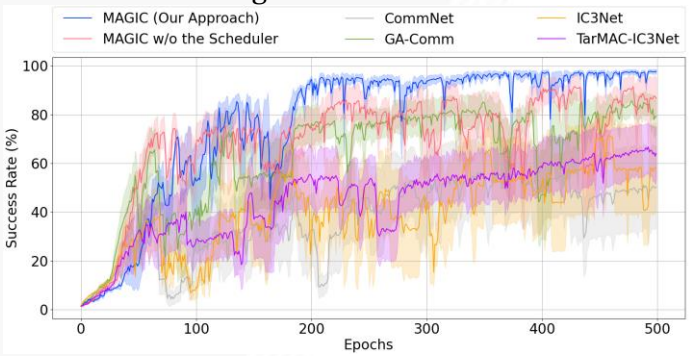
Robotarium



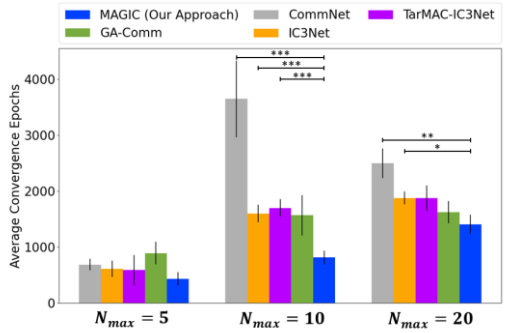
Predator-Prey Communication Heatmaps



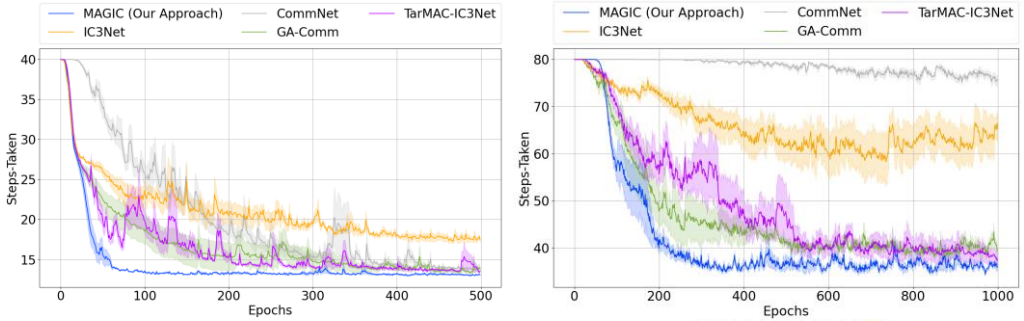
Google Research Football



Traffic Junction



Predator-Prey



Multi-Agent Graph-AttentIon Communication. Yaru Niu, Rohan Paleja, Matthew Gombolay (PI), AAMAS 2021.

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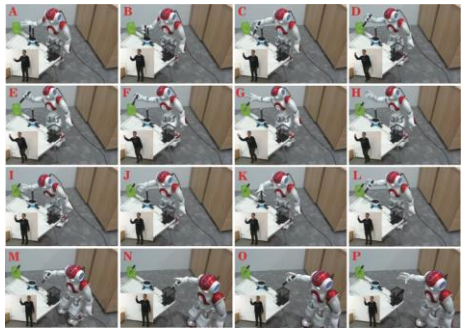
Human-Robot Interaction: Imitation System, Imitation Learning

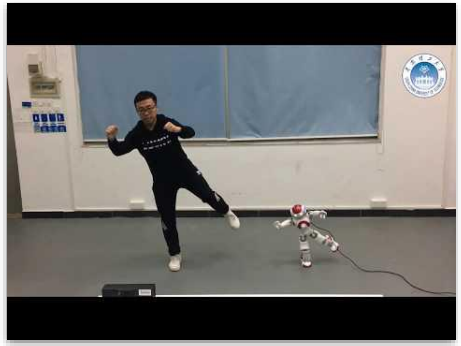
Yaru Niu (https://www.yaruniu.com/)

Whole-body Imitation by Humanoid Robots

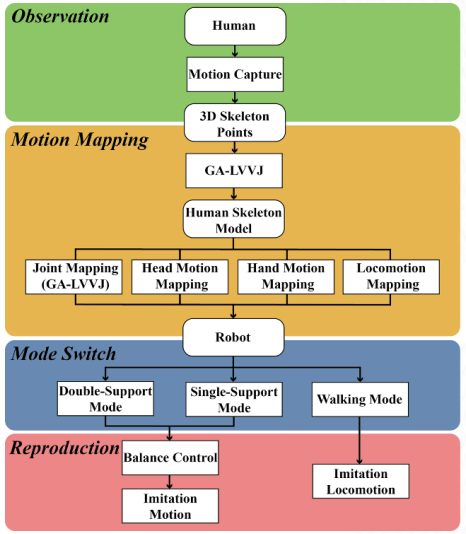
- Built a robot imitation system that enables the imitation of locomotion, and the motions of the head, arms, lower-limbs and hands, without requiring any ancillary handheld or wearable devices or any additional audio or gesture-based instructions.
- Performed a task requiring standing on one foot, grasping and walking by teleoperation in real time using the developed system.
- Endowed the robot with the ability of learning, by designing an imitation learning mechanism with probabilistic models such as Gaussian Mixture Model and Semi-Hidden Markov Model, in which the joint angles of demonstrators were mapped by proposed GA-LVVJ.

Video, click here





Updated short video



Real-time Whole-body Imitation by Humanoid Robots and Tasks-oriented Teleoperation Using an Analytical Mapping Method and Quantitative Evaluation. Yaru Niu*, Zhijun Zhang*(PI), et al., Applied Science (Special Issue Human-Friendly Robotics), 2018.

Active Hierarchical Imitation and Reinforcement Learning (Course Project)

- Developed an Active Hierarchical Imitation and Reinforcement Learning (AHIRL) algorithm which can work in continuous spaces for robot maze navigation. We used DDPG to pre-trained the low-level controller, and employed DAgger to train our high-level controller using human demonstrations. What is more, several active learning methods were designed and used for improving the sample efficiency of the imitation learning in the hierarchical structure.

Video, click here

Active Hierarchical Imitation and Reinforcement Learning. Yaru Niu, Yijun Gu, Zuoxin Tang, arXiv preprint, 2020. (arXiv: 2012.07330)

Human-Robot Interaction: Emotion, Motion Planning

Yaru Niu (https://www.yaruniu.com/)

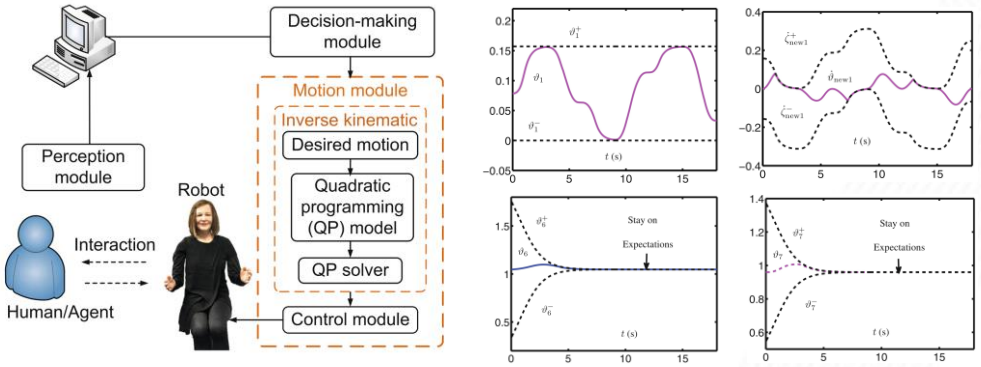
Humanoid Robots' Emotion Expressions by Body Language

- Designed a within-subject experiment in which the participants were asked to choose the type of emotion for different gestures performed by the human and the robot respectively.
- Found that the inherent ambiguity of the gestures and the joint limits can significantly influence the emotion expression of the humanoid robots by body language, through grouping and statistical testing.

Analysis of Influencing Factors on Humanoid Robots' Emotion Expressions by Body Language. Zhijun Zhang (PI), Yaru Niu, et al., International Symposium on Neural Networks (ISNN), 2018.

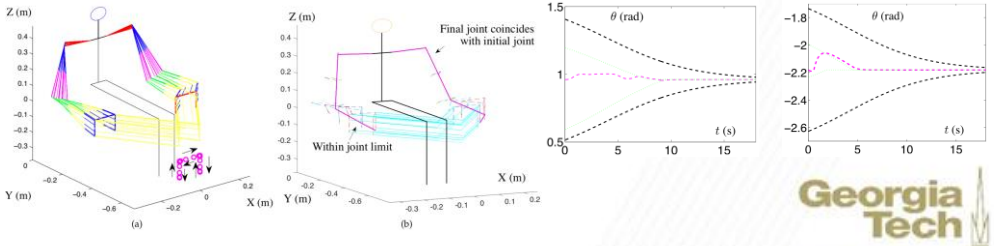
Gesture - Determined Dynamical Schemes for Motion Planning of the Humanoid Robot Arms

- Designed an Time-Varying Gesture-Determined Dynamical (TV-GDD) scheme for motion planning of the redundant manipulators to generate desired gestures, and showed its effectiveness in the experiment based on a humanoid robot .



A Time-Varying-Constrained Motion Generation Scheme for Humanoid Robot Arms. Zhijun Zhang (PI), Lingdong Kong, Yaru Niu, International Symposium on Neural Networks (ISNN), 2018.

- Improved the robustness of TV-GDD by modifying the gesture-determined-dynamic function with consideration of the joint limit margins.



Modification of Gesture-Determined-Dynamic Function with Consideration of Margins for Motion Planning of Humanoid Robots. Zhijun Zhang (PI), Lingdong Kong, Yaru Niu, et al., arXiv preprint, 2020. (arXiv: 2008.06899)

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